MIL-P-52542C(ME)
10 March 1977
SUPERSEDING
MIL-P-52542B(ME)
16 October 1973

MILITARY SPECIFICATION

PACKAGING EQUIPMENT SET: FOR GENERAL SUPPLIES AND SPARE PARTS, MOUNTED ON A 6-TON SEMITRAILER

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the mounting of the components of a packaging equipment set in an M146 semitrailer, and includes the modification of the semitrailer and other necessary fabrication incidental thereto to make a mobile packaging equipment set for general supplies and spare parts, NSN 3540-00-293-9180 (SC 3540-97-CL-E01).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-A-250/1	- Aluminum 1100, Plate and Sheet.
QQ-A-250/2	- Aluminum Alloy 3003, Plate and Sheet.
QQ-A-250/3	- Aluminum Alloy Alclad 2014, Plate and Sheet.

FSC 3540

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

TT-E-485	- Enamel, Semigloss, Rust-Inhibiting.
UU-T-81	- Tags, Shipping and Stock.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-H-3280	- Hoists, Chain, Manually Operated, Packaging of.
MIL-C-12000	- Cable, Cord, and Wire, Electric, Packaging of.
MIL-V-13811	- Varnish, Waterproofing, Electrical, Ignition.
MIL-E-16298	- Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-B-22191	- Barrier Materials, Transparent, Flexible, Heat Sealable.
MIL-D-45322	- Dolly, Trailer Converters.
MIL-T-46780	- Trucks; 2-1/2 Ton, 6x6, Military Design, M44A2, Multi-Fuel Engine Series.
MIL-S-62049	- Semitrailers, Van: For Installation of Special Equipment, Military Design.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

DRAWINGS

ME

D9983-1 and -2	- Preservation and Packaging Equipment Set No. 1 General Assembly.
D9985-1 through -7	- Preservation and Packaging Equipment Set No. 1 Material Handling System.
D9986-1	- Preservation and Packaging Equipment Set No. 1 Venting System.
D9987-1 through -3	- Preservation and Packaging Equipment Set Electrical System.
D9988-1	- Preservation and Packaging Equipment Set No. 1 Fire Fighting System.
D9989-1 through -3, -5 through -8, and -10 through -13	- Preservation and Packaging Equipment Set No. 1 and 2 Preservation Equipment.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASSOCIATION OF AMERICAN RAILROADS

Rules Governing the Loading of Commodities on Open-Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren Street, Chicago, IL 60605).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANS)

C1 National Electrical Code.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

COUNTY OF LOS ANGELES - AIR POLLUTION CONTROL DISTRICT

Rule 66 - Organic Solvents.

(Application for copies should be addressed to the County of Los Angeles - Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)

3. REQUIREMENTS

3.1 Description. The packaging equipment set shall consist of a materials handling system, venting system, electrical system, fire-fighting system, air-supply system, blast cleaning cabinet, storage cabinets, degreaser, infrared drier, wrapping table and dip tanks mounted on or in a van-type semitrailer; and supporting miscellaneous equipment and handtools.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. The contractor is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulative result in improper fits, the contractor shall provide tolerances within those prescribed on the drawings to insure proper fit, assembly, and operation of the packaging equipment set. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish a packaging equipment set for examination and test within the time frame

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

specified (see 6.2), to prove, prior to starting production, that his production methods will produce packaging equipment sets that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examinations and tests as specified.

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more packaging equipment set(s) as specified (see 6.2), for inspection as specified in 4.4.

3.4 Material. Material shall be as shown on the applicable drawings and as specified herein. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product. Where the contact of two dissimilar metals would result in galvanic action, the metals shall be separated by vinyl zinc chromate tape or a coat of primer conforming to TT-E-485 (see 6.9).

3.5 Fabrication and assembly. The general assembly of the semitrailer shall be as shown on Drawings D9983-1 and -2 (including 9989-1 through -3, -5, through -8 and -10 through -13 and as specified herein. The contractor shall fabricate the materials handling system, the venting system, the electrical system, the firefighting system, and the air-supply system and mount them on or in a van-type semitrailer in the positions shown on the applicable drawings. In addition the contractor shall install the items specified in 3.16 and preserve and pack the handtools and miscellaneous equipment as specified in 3.17. The contractor shall make any modification required to mount the equipment in the van-type semitrailer. All systems and installed equipment shall function correctly after installation and after the road test specified in 4.6.2.7.

3.6 Materials handling system. The materials handling system shall be as shown on Drawings D9985-1 through -7 and as specified herein.

NOTE: The van standard lighting system and interior ceiling panels shall be removed to provide for installation of the materials handling system supports. Ceiling panels shall be fitted and reinstalled. The standard lighting system shall not be reinstalled. Sheet aluminum plates shall be used to cover the open areas in the ceiling panels caused by removing the standard lighting system and cutting clearance for installation of the monorail supports. The aluminum plates shall be sized to fit when reinstalling the ceiling panels and shall be secured in place with rivets.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

3.7 Venting system. The venting system shall exhaust air from the semitrailer and shall be as shown on Drawing D9986-1 and as specified herein. Engageable type vents provided on the ducting shall engage the vents provided on the dip tank lids when the lids are in the open position. When tested as specified in 4.6.2.2, the manometer shall show a negative reading.

3.7.1 Duct work. The ducting shall be aluminum conforming to QQ-A-250/1 or QQ-A-250/2, temper best suited for the use. All joints shall be locked with metal strips and shall withstand the road test specified in 4.6.2.7 without damage or loosening.

3.8 Electrical system. The electrical system shall be fabricated and installed as shown on Drawing D9987-1 through -3, and as specified herein. All circuits shall be color coded and correctly identified on the panel box with embossed lettering on pressure sensitive tape. The entire system shall conform to ANS C1 for hazardous locations, Class I, Division 1, Group D. Outlets for handtools shall be such that 110-volt tools cannot be accidentally plugged into 208-volt receptacles or vice versa. Portable electric handtool standard connector plugs shall be replaced with connector plugs that are compatible with and can be plugged into the receptacle outlets furnished by the contractor. Installed equipment individual switches and enclosures containing automatic circuit interrupters shall be approved for operation in ANS C1, Class I, Division 1, Group D locations.

3.8.1 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.8.1.1, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or functioning of the varnish.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

3.8.1.1 Composition. Composition I shall be used unless Composition II must be used in order to comply with local air pollution regulations in the application of the varnish. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that Composition II Material Complies with Rule 66, Air Pollution Control District, County of Los Angeles, CA.

3.9 Firefighting system. The firefighting system shall be as shown on Drawing D9988-1 and as specified herein. All components shall be furnished by the contractor. All joints shall be threaded. Soldered or brazed joints are not permitted. The firefighting system shall be connected to the ventilating fan in such a manner as to deenergize the fan when the system is actuated.

3.10 Air-supply system. The air-supply system shall be as shown on Drawing D9989-13 and shall be equipped with universal quick-connect couplings. The air-supply system shall not leak after installation.

3.11 Reinforcement. Structural reinforcement to the forward bulkhead shall be as shown on Drawing D9983-1 and as specified herein. Material shall be aluminum conforming to QQ-A-250/2 or QQ-A-250/3, temper best suited for the use.

Note: The van forward bulkhead panels shall be removed to provide for installation of the structural reinforcement supports. The bulkhead panels shall be fitted and reinstalled.

3.12 Treatment and painting. All portions of the fabrications and modifications which are normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, Color No. 24533. Mounting brackets or lugs that are fastened to the semitrailer floor shall match the color of the equipment to which they are attached. Exterior parts fabricated and installed by the contractor shall be painted to match the color of the semitrailer. Marred surfaces shall be touched up with the appropriate paints.

3.13 Marking.

3.13.1 Identification marking. The semitrailer shall be identified in accordance with MIL-STD-130. The marking shall be applied to the semitrailer on plates conforming to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material, permanently affixed in a conspicuous location. The plates shall be attached by screws, bolts, or rivets.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

3.13.2 Warning markings.

3.13.2.1 Exhaust system master switch. The following notice shall be stenciled immediately beneath the exhaust system master switch located on the upper portion of the inner forward bulkhead in red block letters not less than 1 inch high on a white background:

"THIS SWITCH MUST BE 'ON'
AT ALL TIMES BEFORE
ALL DOORS ARE CLOSED."

3.13.2.2 Personnel door. The following notice shall be stenciled on the inside of the curbside personnel door immediately below the glass panel in red block letters not less than 1 inch high on a white background:

"EXHAUST SYSTEM MASTER SWITCH
MUST BE 'ON' BEFORE
THIS DOOR IS CLOSED."

3.14 Government-furnished property. The following property in the quantities indicated for each set will be furnished by the Government (see 6.4):

Description	Reference	Quantity Per Set
Semitrailer, M146*	MIL-S-62049	1

* After modification of the M146 as specified in 3.5 through 3.11 the National Stock Number is changed to NSN 2330-00-346-2925 and the item description is changed to read: SEMITRAILER, SPECIAL: 6-Ton; Van Body, for Packaging Equipment (modified ORD M146 Chassis as the listing appears in SC 3540-97-CL-E01).

3.15 Government-loaned property. The Government will loan an M275A2 or equal truck tractor conforming to MIL-T-46780 and an M197A1 Dolly conforming to MIL-D-45322 for use in conducting the road test specified in 4.6.2.7 (see 6.5).

3.16 Installed items. The blast cleaning cabinet, storage cabinets, degreaser, infrared drier, wrapping table, and dip tanks shall be as specified on and installed as shown on Drawing D9989-1.

3.17 Handtools and miscellaneous equipment. Handtools and miscellaneous equipment required for each complete set shall be preserved and packed in the semitrailer as specified in 5.2 and 5.3.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:09Z
Check the source to verify that this is the currentversion before use.

3.18 Workmanship.

3.18.1 Bolted connections. Boltholes shall be accurately punched or drilled and shall have burrs removed. Washers or lockwashers shall be provided, and all bolts, nuts, and screws shall be tight.

3.18.2 Steel fabrication. Steel shall be free from sharp bends and kinks. Straightening shall be done by methods that will not cause injury to the metal. Corners shall be square and true. Flame-cutting may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. Precautions shall be taken to avoid overheating and heated metal shall be allowed to cool slowly. All bends or a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.18.3 Threads. Threads shall be in accordance with Handbook H28.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction packaging equipment set shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

4.3.2 Test. The preproduction packaging equipment set shall be tested as specified in 4.6.2. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3) one or more initial production packaging equipment set(s) will be selected at random by the Government from packaging equipment sets being produced by production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2.1 through 4.6.2.7 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production packaging equipment set shall not exclude the remaining packaging equipment sets from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production packaging equipment set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production packaging equipment set(s) and shall be cause for refusal by the Government to continue acceptance of production packaging equipment set(s) until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on packaging equipment sets previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all packaging equipment sets accepted prior to completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each packaging equipment set shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Test. Each packaging equipment set shall be tested as specified in 4.6.2. Failure of any test shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:09Z
Check the source to verify that this is the currentversion before use.

4.6 Inspection procedure.

4.6.1 Examination. The packaging equipment set shall be examined as specified herein, for the following defects:

101. Material not as specified.
102. Positioning of inlet vents of exhaust system do not correctly engage vapor vents in tank lids.
103. Assembly not complete or not correct.
104. Circuits not color coded or not correctly identified on panel box.
105. Male plugs on electrical handtools not in accordance with specification and could be inadvertently connected with 110 or 220 volt receptacles.
106. Positions of rate-of-rise actuators of the firefighting system not as shown on drawing.
107. Air-supply system pipe and connectors not as specified.
108. Positioning of principal items not as specified on drawings.
109. Treatment and painting not as specified.
110. Identification or other marking incorrect or missing.
111. Handtools or miscellaneous equipment missing.
112. Workmanship not as specified.
113. Data items or provisioning item not furnished (when applicable).

4.6.2 Tests.

4.6.2.1 Materials handling system. The materials handling system shall be operated to assure free movement of all trolleys including those supporting the hoists around all short-radius curves of the monorail. Failure to move freely around the entire track shall constitute failure of this test.

4.6.2.2 Venting system. All vent openings shall be sealed with tape and the exhaust fan shall be operated. The static pressure shall be measured on a manometer. A positive reading on the manometer is an indication of leaks in the ducts or incorrect rotation of the fan and shall constitute failure of the test until the leak has been corrected or rotation of the fan has been reversed. Note: Rotation of the fan may be corrected by reversing the two ungrounded conductors on the generator.

4.6.2.3 Electrical system. The electrical system shall be tested to assure correct operation of all components including installed equipment, lights, and switches; and that all outlet boxes are wired for correct voltage. Failure of any component, light or switch to operate, or incorrect voltage of any of any outlet box, as shown on the drawings shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:09Z
Check the source to verify that this is the currentversion before use.

4.6.2.4 Firefighting system. The firefighting system shall be tested for correct operation. Compressed air may be substituted for the CO_2 tanks. The manual control on the outside bulkhead at curbside shall be operated manually. The rate-of-rise actuators on the ceiling shall be tested by application of a source of heat sufficient to raise the temperature 135° F within 75 seconds. The pressure-operated switch shall be checked during both the above operations to assure that it automatically cuts off the electric power of the ventilating fan at the main switch of the ventilating system and at the same time releases the CO_2. Failure of the manual control or the rate-of-rise actuators to operate the pressure switch to release CO_2 and at the same time to deenergize the fan shall constitute failure of this test.

4.6.2.5 Air-supply system. The air-supply system shall be tested for leaks by the soapsuds test. The suds may be made by using any free-sudsing soap or detergent in warm water. With a positive pressure in the air-supply system, the suds shall be applied by brushing all surfaces of the system. Evidence of leaks indicated by appearance of bubbles shall constitute failure of this test.

4.6.2.6 Installed equipment. All equipment installed by the contractor shall be operated in the semitrailer to determine that it functions correctly. Heating tanks shall be filled to operating level with a thin oil which will drain while cold and can be easily removed with a petroleum solvent, in lieu of the preserving fluid. Any item of installed equipment which does not function correctly shall constitute failure of the test.

4.6.2.7 Roadability. The trailer, with systems, installed and equipment mounted and connected for operation, shall be towed by the Government-loaned M275A2 truck tractor (see 3.15) not less than 20 miles (10 miles with dolly and 10 miles without dolly) over locally available roads at the highest permissible speed, with at least 5 continuous or intermittent miles of road surface as specified herein. "Permissible speed" is defined as the highest safe speed for the combination in the area where the test is being conducted, taking into account road alinement, surface conditions and weather. Road surface shall include one or more of the following types of imperfections ranging in depth from 2 to 8 inches: Pot (chuck) holes, washboard effect (corduroy pattern), ditches, and broken irregular areas. The surface imperfections shall be capable of twisting, jarring, shaking, or vibrating the fabricated systems and installed equipment. The road surface chosen for the test shall be subject to the approval of the contracting officer. During the test the vehicle shall be maneuvered to contact the surface imperfections as often and continuous as practicable. When an acceptable local test road is not available, an approved test area shall be used. The test area shall be subject to the approval of the contracting officer. The combination shall make not less than the specified number of each of the following maneuvers: 10 abrupt stops; 5 abrupt starts;

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

and 20 sharp turns, 10 to the left and 10 to the right. An "abrupt stop" is defined as bringing the combination to a halt from the highest permissible speed within the shortest distance consistent with the velocity of the combination. An "abrupt start" is defined as acceleration of the combination in the lowest forward gear of the truck tractor from a standing position to the maximum permissible speed for that gear within a distance of not more than 150 feet. A "sharp turn" is defined as changing the direction of the combination more than 60 degrees while traveling at the highest safe speed. At the completion of the test run, packaging equipment set shall be retested as specified in 4.6.2.1 through 4.6.2.6 to determine the adequacy of the connections and mountings. All items of installed equipment shall be operated. Deformation or failure of any connection or mounting or failure of any equipment to operate correctly shall constitute failure of this test.

4.7 Inspection comparison. The Government may select packaging equipment sets at any time during the contract production period and subject these packaging equipment sets to the examination specified in 4.6.1 and the tests specified in 4.6.2.1 through 4.6.2.7 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on sets selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production packaging equipment set(s). In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison packaging equipment set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison packaging equipment set(s) and shall be cause for refusal by the Government to continue acceptance of production packaging equipment set(s) until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on packaging equipment sets previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all packaging equipment sets accepted prior to the completion of the inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

4.8 Inspection of packaging.

4.8.1 Preproduction pack inspection.

4.8.1.2 Examination. The preproduction pack shall be examined for the defects specified in 4.8.2.2. Presence of one or more defects shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completely processed set including the miscellaneous equipment and handtools shall be considered a unit of product.

4.8.2.2 Examination. Each complete semitrailer-mounted packaging equipment set shall be examined for the defects marked "X" for the applicable level in Table I.

Table I. Classification of Defects

Major	Defect	Level A	Level B
114.	Materials, methods, and containers not as specified (5.2 and 5.3). Each incorrect material, method, or container shall constitute one defect.	X	X
115.	Disassembly and matchmaking not as specified (see 5.2.1.1 and 5.2.2.1).	X	X
116.	Tags not waterproofed (see 5.2.1.1).	X	-
117.	Preservatives or methods not as specified (see 5.2.1.2).	X	X
118.	Unpainted exterior surfaces not coated with type P-1 preservative when required (see 5.2.1.3 and 5.2.2.2).	X	X
119.	Unpainted ferrous information plates not coated with varnish (see 5.2.1.3).	X	-
120.	Brake systems not preserved as specified in referenced document (see 5.2.1.4 and 5.2.2.3).	X	X
121.	Semitrailer body not preserved as specified in referenced document (see 5.2.1.4 and 5.2.2.3).	X	X
122.	Miscellaneous preservation as specified in referenced document omitted or incorrect (see 5.2.1.4 and 5.2.2.3).	X	X

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

Table I. Classification of Defects (continued)

Major	Defect	Level A	Level B
123.	Inflation of tires not as specified in referenced document (see 5.2.1.4 and 5.2.2.3).	X	X
124.	Trolleys and hoists not anchored to prevent movement on monorail (see 5.2.1.5).	X	X
125.	Door of electrical panel box not securely fastened (see 5.2.1.6.1).	X	X
126.	Main electrical cable not preserved and stored as specified (see 5.2.1.6.2).	X	X
127.	Electric lamps and lamp accessories not preserved as specified (see 5.2.1.6.3 and 5.2.1.6.4).	X	X
128.	Manifold and remote control head of fire-fighting system not removed and preserved as specified (see 5.2.1.7).	X	X
129.	Drains not left open as specified (see 5.2.1.8, 5.2.1.11 and 5.2.1.13).	X	X
130.	Movable bank of infrared drier incompletely blocked (see 5.2.1.9).	X	X
131.	Pipe rollers and paper spools of the wrapping table not stored as specified (see 5.2.1.10).	X	X
132.	Drawers of wrapping table not restrained as specified (see 5.2.1.10).	X	X
133.	Lids of tanks not secured as specified (see 5.2.1.11).	X	X
134.	Doors of cabinets not securely fastened (see 5.2.1.12).	X	X
135.	Lid of degreaser not secured as specified (see 5.2.1.13).	X	X
136.	Nozzles, air jets, and tip covers of blast cleaning cabinet not individually preserved and stored as specified (see 5.2.1.14).	X	X
137.	Door of blast cleaning cabinet not secured (see 5.2.1.14).	X	X
138.	Rotating electrical equipment not preserved as specified (see 5.2.1.15).	X	X
139.	Electrical components not preserved as specified in referenced document when not removed (see 5.2.1.15).	X	X

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

Table I. Classification of Defects (continued)

Major	Defect	Level A	Level B
140.	Carpenter equipment set and master mechanic's tool set not preserved in accordance with referenced document (see 5.2.1.16).	X	X
141.	Lids of chests not securely fastened (see 5.2.1.16).	X	X
142.	Fire extinguishers not coated with preservative when required (see 5.2.1.17).	X	X
143.	Fire extinguishers not preserved as specified (see 5.2.1.17).	X	X
144.	Electric handtools not preserved as specified (see 5.2.1.18 and 5.2.2.4).	X	X
145.	Miscellaneous handtools not preserved as specified (see 5.2.1.19).	X	X
146.	Handtools not stored as specified (see 5.2.1.18 and 5.2.1.19).	X	X
147.	Miscellaneous supplies not preserved as specified (see 5.2.1.20 through 5.2.1.30).	X	X
148.	Miscellaneous supplies not stored as specified (see 5.2.1.20 through 5.2.1.25 and 5.2.1.28 through 5.2.1.30).	X	X
149.	Consolidation not as specified (see 5.2.1.31).	X	X
150.	Inadequate cushioning, blocking, and bracing to prevent movement of the contents in the consolidation containers (see 5.2.1.31).	X	X
151.	Consolidation containers exceeding the weight limitation or of a size and shape that will not fit into available space (see 5.2.1.31).	X	X
152.	Consolidation containers and components not blocked, braced and anchored as specified (see 5.3.1).	X	X
153.	Marking illegible, incomplete, incorrect, or missing (see 5.4).	X	X

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination within the time frame specified (see 6.2), to prove prior to producing production packs that his applied preservation, packing,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

and marking comply with the packaging requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.6). The preproduction pack may be accomplished utilizing either the preproduction packaging equipment set or a production packaging equipment set. If the preproduction set is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the preproduction packaging equipment set and production packaging equipment sets.

5.2 Preservation. Preservation shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly and matchmarking. Disaasembly shall be limited to that specified herein, all of which can be performed without special tools. Removed parts and assemblies shall be matchmarked to facilitate reassembly. Matchmarking shall be on shipping tags conforming to UU-T-81, Type A, or on metal tags. The cloth tags and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

5.2.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116. Methods of preservation indicated by symbols herein shall conform to the corresponding symbols in MIL-P-116.

5.2.1.3 Unpainted exterior surfaces. Unpainted exterior metal surfaces of components requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative. Unpainted ferrous metal identification and instruction plates shall be coated with varnish conforming to MIL-V-13811.

5.2.1.4 Semitrailer. The semitrailer components shall be preserved in accordance with MIL-STD-281, Level A (mobile), as applicable.

5.2.1.5 Material-handling system. The chain hoists shall be preserved in accordance with MIL-H-3280, Level A. The trolleys and hoists shall be securely anchored to prevent movement on the monorail during transit and storage.

5.2.1.6 Electrical system.

5.2.1.6.1 Panel box. The door of the electrical panel box shall be securely fastened to prevent accidental opening during transit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

5.2.1.6.2 Main electrical cable. The main electrical cable shall be coiled in an elongated coil that will lay on the floor in the aisle to the rear of the infrared drier and shall be preserved with ends wrapped, as specified for coils in MIL-C-12000, Level A, except that waterproof wrapping of the coil shall not be required.

5.2.1.6.3 Incandescent lamps. Incandescent lamps shall be removed from all fixtures and individually preserved and cushioned in commercial-type containers. The preserved lamps shall be consolidated, like lamps together in snug-fitting boxes conforming to PPP-B-636, W5c, style optional.

5.2.1.6.4 Extension lights. The cords of extension lights shall be coiled and restrained with tape and the lamps shall be individually preserved in boxes conforming to PPP-B-566, style, type, and class optional, as appropriate.

5.2.1.7 Firefighting system. The system shall be disconnected at the carbon dioxide cylinders and the cylinder cap shall be installed fingertight. No preservative shall be applied to interior of piping or the valves. The manifold and the remote control head shall be removed and individually preserved in boxes conforming to PPP-B-566, style, type, and class optional, as appropriate.

5.2.1.8 Air lines and drains. Interior surfaces of air lines and drain valves shall be fogged with MIL-L-21260, Type I, Grade 10 preservatives, and the drains shall be left open.

5.2.1.9 Infrared drier. The movable bank of the infrared drier shall be secured in the retracted position and blocked at the floor level by a wood 2 by 4 extending to the curbside wall of the semitrailer. The door of the drier cabinet shall be secured to prevent accidental opening during transit.

5.2.1.10 Wrapping table. The pipe rollers and paper spools shall be removed and the pipe rollers shall be wrapped together and secured. The paper rollers shall be placed in the table drawers and cushioned to prevent movement. The drawers shall be restrained by the metal rod provided for the purpose.

5.2.1.11 Tanks. All cleaning and preservative tanks shall be drained, cleaned, and dried. Interior unpainted ferrous surfaces shall be coated with MIL-L-21260, Type I, Grade 10 preservative, and the drains shall be left open. The lids shall be secured in the closed position with tape conforming to PPP-T-60, Type IV.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:09Z
Check the source to verify that this is the currentversion before use.

5.2.1.12 Storage cabinets. Items stored in the storage cabinets shall be securely blocked, braced, and anchored to prevent movement within the cabinet and the doors of the cabinet shall be securely fastened to prevent opening during transit.

5.2.1.13 Degreaser. The degreaser shall be preserved as specified in 5.2.1.11 for cleaning and preservative tanks.

5.2.1.14 Blast-cleaning cabinet. Unprotected ferrous metal surfaces shall be coated with Type P-1 preservative. The light bulb shall be preserved as specified in 5.2.1.6.3. Blast nozzles, air jets, and tip covers shall be individually preserved Method IC-1 in transparent bags and the bags shall be heat sealed. All preserved components shall be placed in the blasting compartment and cushioned to prevent damage and the door shall be closed and secured.

5.2.1.15 Rotating electrical equipment and components. Electrical equipment having rotating parts such as motors, fans, and blowers shall be preserved in accordance with MIL-E-16298, Level A, using the alternate method therein for equipment assembled to machines.

5.2.1.16 Carpenter's equipment set and master mechanic's tool set. The tools of these sets shall be preserved in accordance with PPP-P-40, Level A. Tools not specifically identified herein shall be preserved as specified for tools of similar design. The tools shall be placed in the chests and the lids shall be closed and secured.

5.2.1.17 Fire extinguishers. Unprotected metal surfaces of the fire extinguishers requiring a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative and preserved in boxes conforming to PPP-B-636, W5c or W6c, style optional. The preserved extinguishers shall be placed in the storage cabinet and braced, blocked, and anchored to prevent movement.

5.2.1.18 Electric handtools. Portable electric handtools shall be preserved Method II-B and placed in storage cabinet.

5.2.1.19 Miscellaneous handtools. Handtools shall be preserved in accordance with PPP-P-40, Level A, and placed in the storage cabinet.

5.2.1.20 Aprons, rubber gloves, and abrasive cloth. Each apron, each pair of rubber gloves, and each sleeve of abrasive cloth shall be individually preserved Method IC-1, without preservative, in heat-sealed bags of transparent material conforming to MIL-B-22191, Type II, and placed in one of the storage cabinets.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:09Z
Check the source to verify that this is the current version before use.

5.2.1.21 Brushes (paint, varnish, stencil, and marking). The paint, varnish, and stencil brushes shall be individually preserved and the marking brushes of like description shall be preserved together Method IC-1, without preservative, in heat-sealed bags of transparent material conforming to MIL-B-22191, Type II, and placed in the wrapping table drawers.

5.2.1.22 Scrub brushes. The scrub brushes shall be preserved together, without preservative, in a box conforming to PPP-B-566, style, type, and class optional, and placed in the cabinet under the infrared drier.

5.2.1.23 Wire brushes. The wire shall be coated with Type P-1 preservative and the brushes shall be individually preserved in boxes conforming to PPP-B-566, style, type, and class optional, and placed in the cabinet under the infrared drier.

5.2.1.24 Flashlights and thermometers. Flashlights and thermometers in the commercial cartons shall be preserved Method IC-1, without preservative, in heat-sealed bags of transparent material conforming to MIL-B-22191, Type II, and placed in a drawer of the wrapping table.

5.2.1.25 Staplers, moisteners, sprayers, staples, brass stencils. Each stapler, moistener, and sprayer, each box of staples, and each set of stencils in the commercial cartons shall be preserved in accordance with Method III and placed in the storage cabinet.

5.2.1.26 Strapping machines. Strapping machines shall be preserved in accordance with Method I, using MIL-L-21260, Type I, Grade 10 preservative, and wrapped in barrier material conforming to MIL-B-121, Type I, Grade A, Class 2.

5.2.1.27 Flat steel strapping. Flat steel strapping in the original, paper-wrapped, commercial coil shall not require packaging.

5.2.1.28 Funnels and galvanized buckets. Funnels and buckets shall be preserved in accordance with Method III and placed in the cabinet under the infrared drier.

5.2.1.29 Steel wire. The galvanized steel wire in the commercial secured coil shall be preserved in accordance with Method III and placed in the cabinet under the infrared drier.

5.2.1.30 Data and provisioning items.

5.2.1.30.1 Data items. Data items shall be preserved Method IC-1 and placed in the top drawer of the wrapping table.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:09Z
Check the source to verify that this is the current version before use.

5.2.1.30.2 Provisioning items. The preservative application criteria and methods of preservation of MIL-P-116 shall be used to preserve and package the provisioning items.

5.2.1.31 Consolidation. Components and supplies not placed in drawers, cabinets, and compartments shall be consolidated in quantities not to exceed 200 pounds gross weight in snug-fitting boxes conforming to PPP-B-601, Domestic Type, style optional. Boxes shall be of a size that will fit available spaces in the semitrailer and to accomplish the packing specified herein. Strapping shall not be required. Blocking, bracing, and anchoring in accordance with MIL-STD-1186 shall be applied as required to prevent movement of the items within the boxes.

5.2.2 Level B. Each complete semitrailer-mounted packaging equipment set shall be preserved as specified for Level A with the following exceptions.

5.2.2.1 Disassembly and matchmaking. Disassembly and matchmaking shall be as specified in 5.2.1.1 for Level A, except tags used for matchmaking need not be waterproofed.

5.2.2.2 Unpainted exterior surfaces. Unpainted exterior surfaces shall be coated with preservative as specified in 5.2.1.3 for Level A, except identification and instruction plates need not be coated with varnish.

5.2.2.3 Semitrailer. The semitrailer components shall be preserved in accordance with MIL-STD-281, Level B (mobile).

5.2.2.4 Portable electric handtools. Portable electric handtools shall be preserved Method III.

5.2.3 Commercial. The components of the semitrailer-mounted packaging equipment set shall be preserved to afford protection against deterioration and damage during shipment and handling from the contractor to the initial destination.

5.3 Packing. Packing shall be Level A or Commercial as specified (see 6.2).

5.3.1 Level A. Each semitrailer-mounted packaging equipment set preserved and prepared for shipment in accordance with this specification shall be shipped without packing. Consolidation containers and components shall be positioned within the semitrailer and blocked, braced, and anchored in accordance with MIL-STD-1186 to prevent movement within the semitrailer. When shipped by rail, the loading on the car shall be in accordance with the applicable requirements of the Association of American Railroads, "Rules Governing the Loading of Commodities on Open-Top Cars."

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:09Z
Check the source to verify that this is the current version before use.

5.3.2 Commercial. Each complete semitrailer-mounted packaging equipment set shall be prepared for mobile delivery in compliance with applicable carrier rules and regulations.

5.4 Marking.

5.4.1 Military packaging. In addition to any special marking required by the contract or order, marking shall be in accordance with MIL-STD-129.

5.4.2 Commercial packaging. In addition to any special marking required by the contract or order, marking shall reflect as a minimum, the item name, National Stock Number, quantity, contract or purchase order number, and appropriate address markings. The marking shall be applied with a waterproof material, be legible, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The semitrailer-mounted packaging equipment set is intended to be used in the field, or in areas not served by permanent installations, for packaging of general supplies and repair parts whose individual weight does not exceed 500 pounds.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2.1).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(d) When initial production inspection is required and the number of packaging equipment sets to be furnished, when applicable (see 3.3).
(e) Time frame required for submission of preproduction pack (see 5.1).
(f) Degree of preservation and degree of packing required (see 5.2 and 5.3).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:09Z
Check the source to verify that this is the current version before use.

6.3 Preproduction model. Any changes or deviations of production packaging equipment sets from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish packaging equipment sets conforming to this specification.

6.4 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.14.

6.5 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.15.

6.6 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the packaging equipment sets in accordance with this specification.

6.7 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists and contractor's maintenance and operation manuals to be furnished with each packaging equipment set.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions on shipment of packaging equipment sets.

6.9 It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 3540-A121

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:09Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-P-52542C(ME) Packaging Equipment Set: For General Supplies and Spare Parts, Mounted on a 6-Ton Semitrailer

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:09Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Mobility Equipment Research and
Development Command
ATTN: DRDME-DS
Ft. Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.